Project: A Reverse-Hoops-Playing Robot Arm

Robotics (B-KUL-H02A4A)

2026

1 Introduction

In this project you will focus on motion planning and control of a robot arm with a hoops ring as an end-effector. The simulation environment is depicted in Figure 1. In the simulation, a ball will be launched at the robot, and the robot should move the hoops ring such that the throw scores.

2 What is provided?

The following is provided to help you complete the project:

1. A basic simulation environment for the platform, as seen in Figure 1.
2. A URDF file defining the geometry of the robot and the environment, including the definition of the tool center point frame `{tcp}`
3. A Python notebook `hoops.ipynb` that demonstrates how to interact with the simulation environment and provides a starting point for the project.

3 Tasks

You will move the hoops ring via acceleration control of the robot joints. The URDF file defines position and velocity limits of the robot's joints, and acceleration limits are given as $2\,\mathrm{rad\,s^{-2}}$ for each joint. The URDF file also defines a tool center point

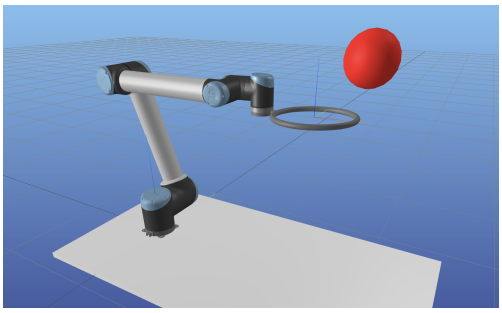

Figure 1: Simulation.

at the center of the ring. The hoop ring has an inner radius of 150 mm, and the ball has a radius of 120 mm. The setup measures the pose of the tool center point and the current position of the ball.

1. Implement a prediction of the ball's trajectory from measurements of its position. Choose one of the filters for state estimation you know from the class and motivate your choice.

2. Use a simple strategy to determine a point on the ball's predicted trajectory at which the robot should catch the ball.

3. Compute the required motion to catch the ball with the robot arm with an optimization-based controller. In order to not get an imaginary net of the hoops ring tangled up, the ring may never face with its top side towards the ground. Avoid self-collisions and collisions with the table in the motion plan. Respect the joint limits on position, velocity, and acceleration.

4. Implement a smarter strategy to determine the interception point for higher success rate and shorter interception time.

4 Assumptions

1. We disregard collision effects. As long as a ball passes through the hoops ring without touching it, the catch is assumed to be successful.

2. There is independent and normally distributed noise on the ball position measurements with zero mean and a standard deviation of 1 mm.

3. There is no real-time requirement on your algorithms; you can disregard the required computation time.